

AYE
TECH HUB

• AI & PRODUCTIVITY • LESSON 06

Google Gemini AI

Models · Workspace · NotebookLM · 1M Token Context

Lesson 06 of 30 | AI & Productivity Program

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- Gemini Flash, Pro and Ultra — which model for which task
- The 1-million token context window and what it makes possible
- Gemini in Google Docs, Sheets, Gmail and Slides — step by step
- NotebookLM: AI research assistant trained on your own documents
- 10 practical Gemini workflows you can use in engineering today

01 | What Is Google Gemini?

Gemini is **Google's flagship AI model family**, developed by Google DeepMind. It was designed from the ground up to be multimodal — understanding and generating text, images, audio, video, and code in a single model. Unlike earlier AI systems that were retrofitted for multiple modalities, Gemini was built multimodal from day one.

Gemini's most remarkable feature is its **context window**. Where GPT-4o can process ~96,000 words and Claude can process ~150,000 words, Gemini 1.5 Pro can process **1 million tokens — roughly 750,000 words, 700 pages of text, or 11 hours of audio**. This changes what is possible with AI.



Figure 1 — Gemini Model Tiers: Flash / Pro / Ultra

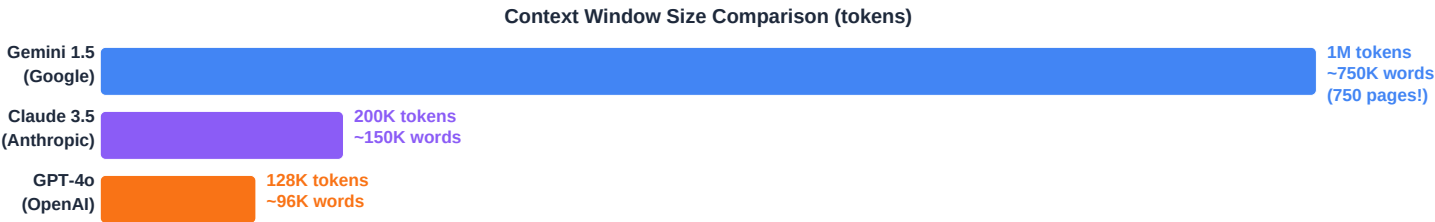


Figure 2 — AI Context Window Comparison: GPT-4o vs Claude vs Gemini

Feature	Gemini Flash	Gemini Pro (Advanced)	Gemini Ultra (Enterprise)
Context window	1M tokens	1M tokens	1M tokens
Speed	Fastest	Balanced	Slower (most capable)
Image understanding	Yes	Yes	Yes (best quality)
Video understanding	Yes	Yes	Yes
Code generation	Good	Very good	Excellent
Google Workspace	Partial	Full (Advanced)	API / Enterprise
Free tier	Yes (generous)	Gemini Advanced (paid)	API pricing only
Best for	Volume, speed, free use	Daily professional use	Research, enterprise

02 | Gemini in Google Workspace

If you use Gmail, Google Docs, Sheets, or Slides, you already have access to Gemini — it is integrated directly into every Google Workspace product. **Gemini Advanced** (paid) unlocks the full 1M context window and most powerful features. The free tier provides basic Gemini capabilities.

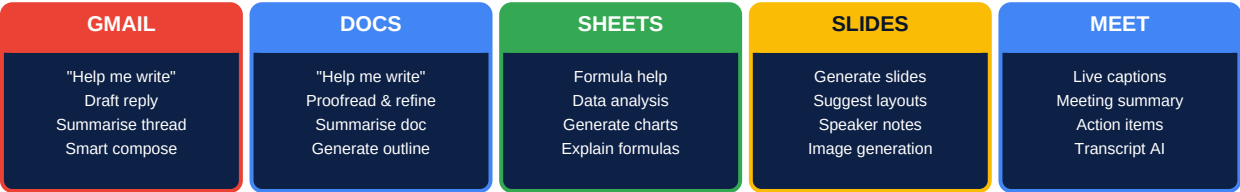


Figure 3 — Gemini AI Features Across Google Workspace

Workspace App	How to Access Gemini	Most Useful Features
Gmail	Click the "Help me write" pencil icon when composing, or "Summarise this email" in inbox	Draft professional replies in seconds; summarise long email threads; smart compose suggestions
Google Docs	"Help me write" sidebar → describe what you need; or select text → ask Gemini to refine	Generate first drafts; proofread and improve tone; summarise long documents; create outlines
Google Sheets	Click Gemini icon → describe what you want in natural language	Explain complex formulas; generate formulas from description; analyse data; create charts
Google Slides	"Help me create a presentation" → enter topic → Gemini generates full slide deck	Generate slides from topic or notes; suggest designs; write speaker notes per slide
Google Meet	Automatic (no activation needed for paid accounts)	Live captions; post-meeting summary with action items; searchable transcript

Engineering Use Case — Gemini in Sheets

Paste your solar system calculation data into Google Sheets. Then ask Gemini: "Analyse this data and identify the three biggest inefficiencies. Create a chart showing panel output vs temperature. Write a formula to calculate the system yield for each month." Gemini does all three — no formula knowledge needed.

03 | NotebookLM — AI Research on Your Own Documents

NotebookLM (notebooklm.google.com) is a completely different kind of AI tool. Instead of answering from its training data, NotebookLM is trained exclusively on the documents you upload. It only answers from your sources — and always cites which source it drew from. This eliminates hallucination for your specific content.

Feature	What It Does	Engineering Application
Source-grounded answers	Only uses information from your uploaded PDFs, docs, and URLs. Every answer has a source citation.	Upload IEC standards, project specs, technical manuals — ask questions and get cited answers
Audio Overview	Generates a podcast-style audio discussion between two AI hosts about your documents	Convert a 200-page technical report into a 20-minute audio summary for learning on the go
Study Guide generation	Creates quizzes, flashcards, and summaries from your sources automatically	Upload textbooks or lecture notes → auto-generate exam practice questions
Timeline / FAQ generation	Extracts and organises key events, Q&A; pairs, and key terms from your sources	Upload project documentation → generate FAQ for client presentations or new team members
Multi-source synthesis	Analyses and connects information across up to 50 uploaded sources simultaneously	Upload multiple project reports, compare findings, identify contradictions and gaps

04 | 10 Practical Gemini Workflows for Engineers

#	Workflow	How to Do It
1	Understand a long technical standard	Upload 200-page IEC/ISO standard to NotebookLM. Ask: "What are all requirements for [topic]?"
2	Write a technical proposal	Gemini in Docs: "Write a 3-page technical proposal for a 50 kW solar system for a commercial building."
3	Analyse project data	Paste data into Sheets → ask Gemini: "Identify trends and flag any values outside normal range."
4	Create a presentation instantly	Slides → "Help me create" → "Solar PV system design for a factory roof — 10 slides, technical audience"
5	Process a long engineering report	Upload PDF to NotebookLM → "Summarise key findings, list all recommendations, highlight cost figures."
6	Draft a client email	Gmail → "Help me write" → "Professional email to client explaining 2-week project delay due to equipment."
7	Formula explanation (Sheets)	Select a complex formula in Sheets → Gemini sidebar → "Explain this formula step by step."
8	Compare multiple standards	Upload EC2 and ACI 318 to NotebookLM → "Compare the load combination requirements in both standards."
9	Generate a meeting agenda	Docs → "Help me write" → "Agenda for project kickoff meeting with client, contractor, and MEP consultant."
10	Audio briefing from documents	NotebookLM → upload reports → Audio Overview → listen while commuting to learn project context.

What comes next — AI-007

NotebookLM for Research — a deep dive into using NotebookLM as your personal AI research assistant: advanced source management, the inline citation system, building a knowledge base, and the Audio Overview feature for learning on the go.